



**UTM**  
UNIVERSITI TEKNOLOGI MALAYSIA

## **DIGITAL LOGIC**

**(SECR1013-03)**

### **LIFT ELECTRONIC CONTROLLER SYSTEM PROJECT**

**Lecturer:**

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## **Dedication & Acknowledgement**

Throughout doing this lift electronic controller system project and writing of this report we have received a great deal of support and assistance.

First of all, we would like to express our deepest gratitude to our lecturer , Dr. Nurfazrina binti Mohd Zamry whose expertise and guidance have been instrumental in the completion of this project. Thank you for imparting your knowledge of Digital Logic this semester and showing us how to apply it in real-life applications. Your patience has enriched our understanding and enhanced the quality of our work.

I would like to extend my sincere thanks to my group members who did the project together. Without teamworks, we would not have done this project successfully and efficiently. Each member had brought unique skills, perspectives, and enthusiasm to this project. Thank you for your dedication, hard work, and the seamless coordination that made this project a success.

Additionally, we want to express our appreciation to any other individuals, friends, or family members who have supported us during this period. Your encouragement and understanding during the challenging moments were invaluable to us.

Lastly, this project has been a learning experience filled with growth, challenges, and achievement, and we are sincerely grateful for everyone who played a part in its realization.

## The Background

In this project, we have implemented a comprehensive lift electronic controller system, which will bring us to the level we want to go. The full circuit is designed and drawn by using Deeds.

To design the full circuit, we have to use combinational circuit and sequential circuit. The components we had used for the *combinational circuit* are **Switches, Decoder, Basic gates, LEDs, 3-bit Comparator , 7-segment display** and **Arithmetic Comparator**. Whereas the components we had used for the *sequential circuit* are **3-bit up/down counter** and **clock disabler**.

In this project, there are a few enhanced features that we have add on which are LED lights which show the lift is fuctioning, 4-digit password, emergency stop, and maximum weight limit .

## **Problem & Solution**

In developing a successful lift electronic controller system, it was critical to identify the key features that a strong system should include.

First and foremost, precision in reaching the intended destination was an important consideration. To address this, we had implemented a circuit that resulted in a consistent and accurate arrival at the desired floor. This improvement was critical in delivering a consistent and efficient transportation experience to users.

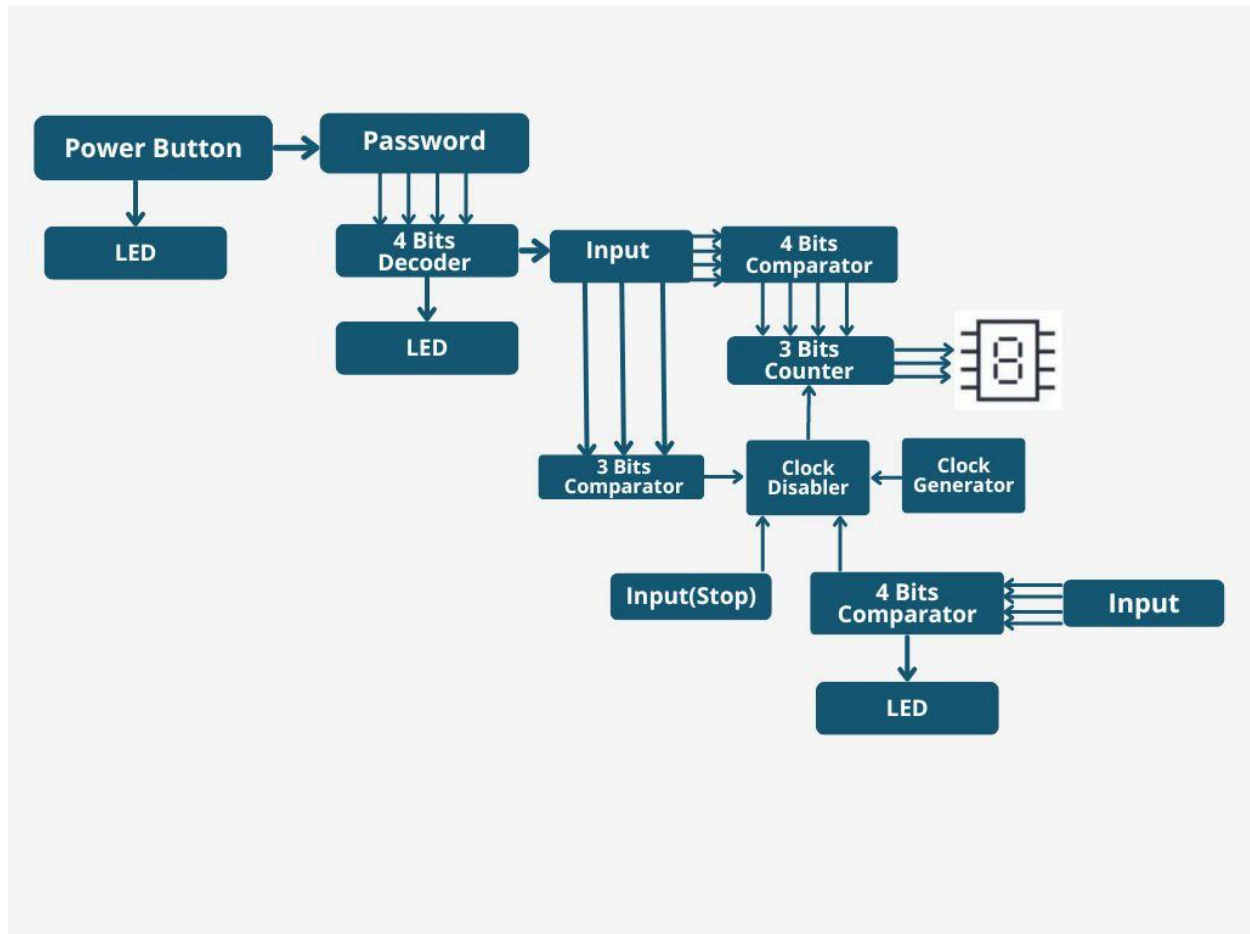
Besides that, another top priority that emerged was ensuring passenger safety. In response to this concern, we added an emergency stop feature to the system. This feature allows for an immediate halt in lift operations in the event of unexpected events, contributing to a quick and secure response to emergencies and prioritizing passenger safety.

Security was also a priority, and to prevent unauthorized access to the lift, we implemented a password protection system. This additional layer of security ensures that only authorized individuals can operate the elevator, reducing the risk of unauthorized use while also improving the overall safety and security of the lift system.

Moreover, to ensure that the lift can function properly, we have added a maximum weight limit. When the lift exceeds the maximum weight limit, an alarm sounds, alerting individuals to the situation. This noticeable warning is complemented by a visible alarm light from our dedicated circuit below, providing a comprehensive safety mechanism.

By addressing these aspects, our aim was to create a comprehensive Lift Electronic Controller System that not only meets basic functionality requirements but also prioritizes safety, security, and precision in operation.

## Block Diagram



**Fig 1: Block Diagram**

## **The Requirements**

### **Input switches**

The user can enter the level of floors they wish to reach using one input switch.

### **Decoder**

The 4-bit decoder is used to decode the 4 digit binary password which is entered by the user using the input switches and the power button remains on for the decoder to be operational.

### **Clock disabler**

The clock disabler is only active when all five inputs are high, and it is utilized to start and stop the counter operation when the level of floors is reached.

### **3-bit count up/down counter**

The counter begins and ends counting based on the clock enabler and the valid password entered by the user.

### **7-Segment Display**

The 7-Segment display shows the current levels of floor reached.

### **3 bit-Comparator**

The 3 bit-comparator is used to compare the value of the current levels of floor and the level of the floors the user wishes to reach.

### **4 bit-Comparator**

The function of the 4-bit comparator is to serve as a 4-bit synchronous counter.

### **Power button**

This input switch is essential as it allows the user to choose to use the lift or not.

The decoder will only operate when the power button is active.

### **LED**

The output of the decoder and the comparator is connected to an LED. When the user enters the correct password or the weight limit is exceeded, the LED will light up.

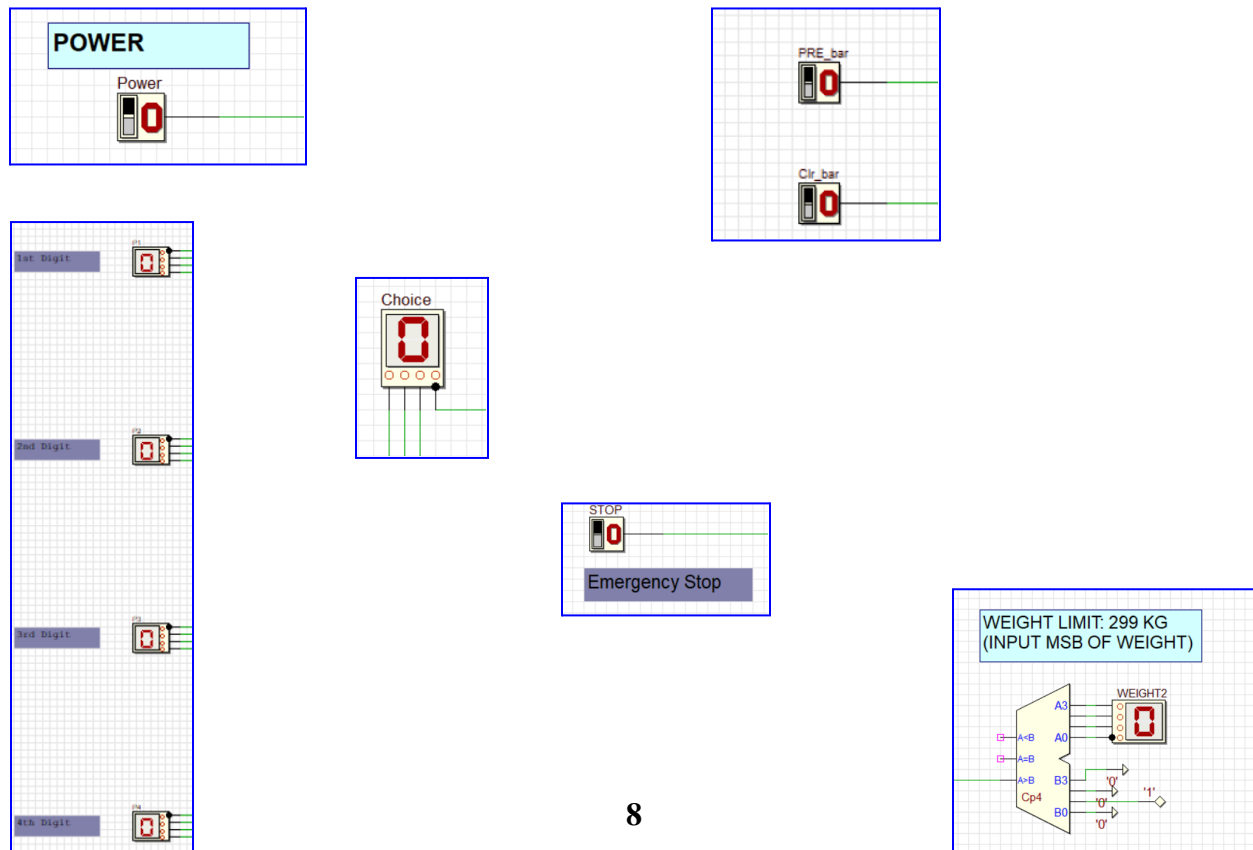
### **Preset & Clear button**

These two buttons are used to the initial display value of the current floor.

# System Implementation

## 1. Input Switch

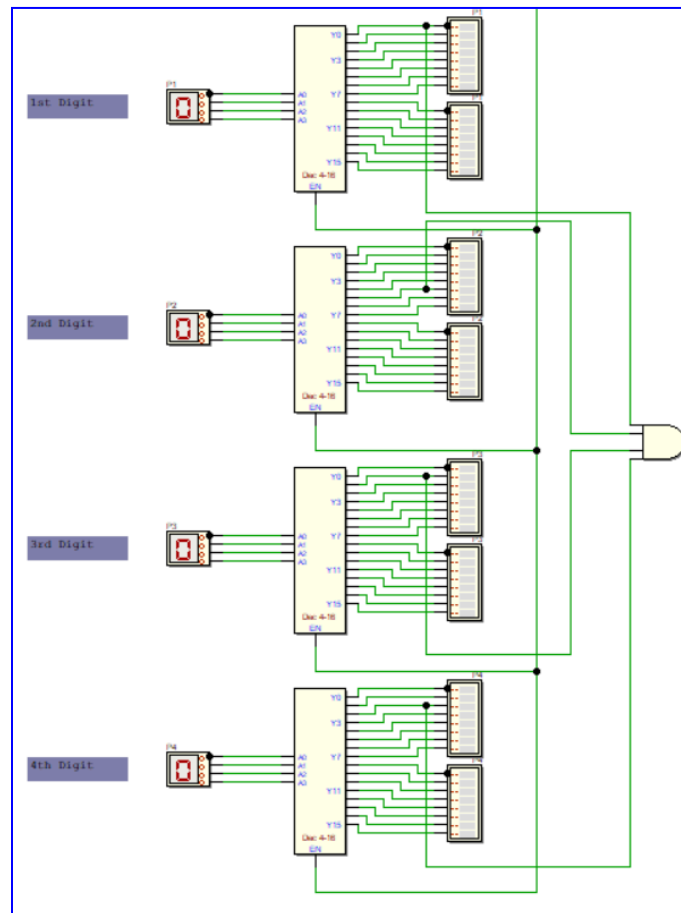
Initially, an input switch serves as the power button (Power) for starting and stopping the lift. Following that, Pre\_bar and Clr\_bar input buttons are utilized to govern the initial display value of the current floor. Activating Pre\_bar initializes the floor to 0, while Clr\_bar must be activated subsequently to activate the circuit. A 4-input Hex Digit (4 bit) switch (P1, P2, P3, P4) allows the user to input the password. Upon successful verification of the password, an LED illuminates to indicate that operations can proceed. Additionally, a weight sensor is integrated into the circuit, represented by a 4-input Hex Digit (4 bit) switch. The most significant bit (MSB) of the switch is employed, and if the WEIGHT2 switch receives a value greater than 2 for the MSB, an LED signals an overweight situation. An emergency stop input switch is incorporated, when activated, it promptly stops the lift. Finally, another 4-input Hex Digit (4 bit) switch enables users to select their desired floor. Notably, there's a limitation on the chosen floor, allowing input only within the range of 0-7. If a floor higher than 7 is selected, the lift remains non-operational until a floor within the valid range is chosen.





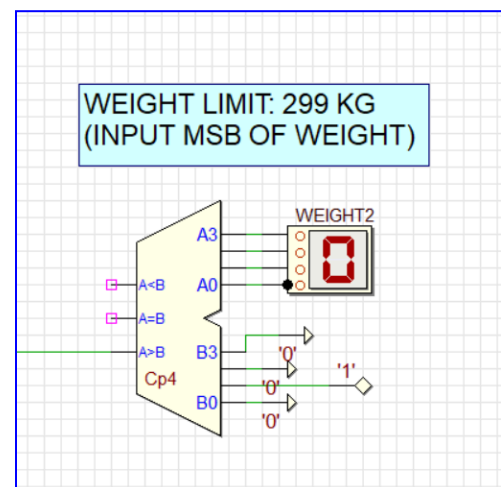
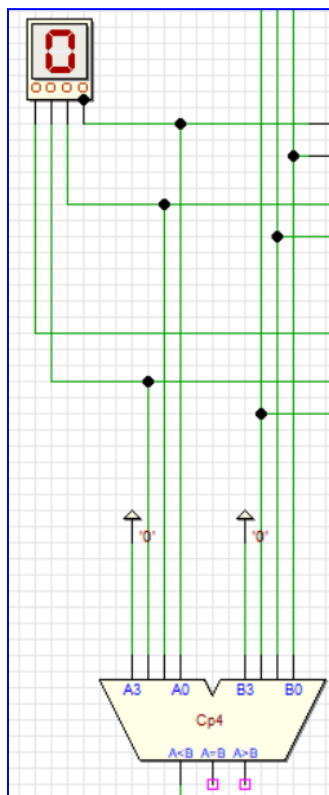
## 2. Decoder

We want to set a 4-digit password. So, we use four 4-bit decoders to decode the value of the user input using the Hex Digit (4 bit) input switch. Each decoder represents each digit of the password. There is a power button input switch to select either starting or stopping the lift. The enable pin of the 4-bit decoder is connected to the power button. It means that the decoder will only activate when the power button is activated. The output of each decoder is connected to two LED arrays to represent 16 different states. It will light up when the user has selected the input value ranging from 0 to F for each decoder. The output of the decoders is connected to an AND gate, and the AND gate is connected to the LED. Once the LED lights up, it means the correct password is input by the user, and the lift is able to operate for the user. The output of the decoder is also connected to (5 inputs) AND gate with the comparator, a NOT gate from a MSB digit of input switch (Choice), clock generator and a NOR gate of emergency stop and weight sensor. When all the outputs from the decoders are high, it will activate the clock enabler.



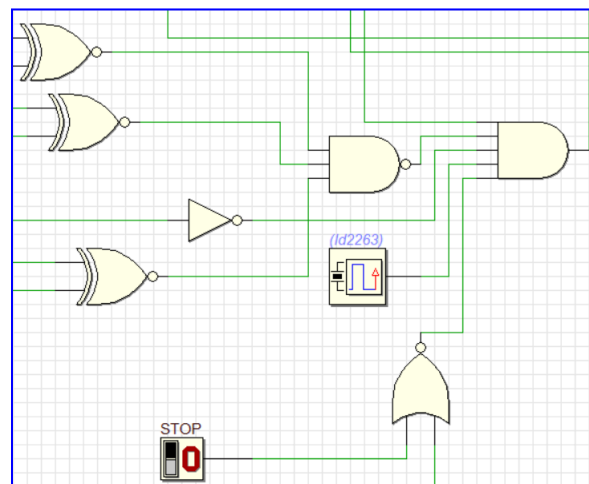
### 3. Arithmetic Comparator (4-bit Comparator)

For this circuit, we employ two arithmetic comparators. The first comparator serves to govern the 4-bit synchronous counter, allowing it to count up (0) or count down (1). If the selected floor (A) is higher than the current floor (B), the output (A<B) will register as 0, prompting the counter to ascend until it reaches the chosen floor. Conversely, if the current floor is higher than the user-selected floor, the output (A<B) will be 1, instructing the counter to descend to the desired floor. The second arithmetic comparator is tasked with determining whether the current weight exceeds the designated limit (299 kg). This comparator focuses on the Most Significant Bit (MSB) digit for comparison. The B value is preset at 0010 (2), while the A value is derived from the weight sensor. However, in this setup, we utilize a Hex Digit input switch to represent the MSB digit from the sensor. Should the sensor transmit a value greater than 2 to the comparator, the output A>B will go high (1), triggering an Alarm LED to indicate an overweight situation.



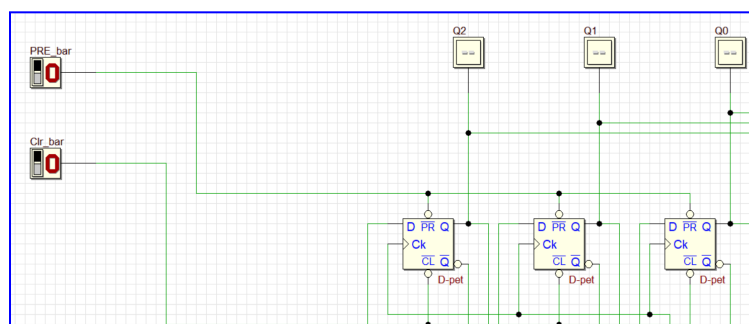
#### 4. Clock Disabler

The Clock Disabler is configured through a 5-input AND gate, where the inputs include the clock source, signals from the password detector, comparator, emergency stop/overweight signal, and input validator. Activation of the clock disabler occurs only when all five inputs register as high. The output from the 5-input AND gate is then linked to the clock input of the counter's flip-flop. This connection serves to start and halt the counter's operation when both the current floor and chosen floor conditions are satisfied.



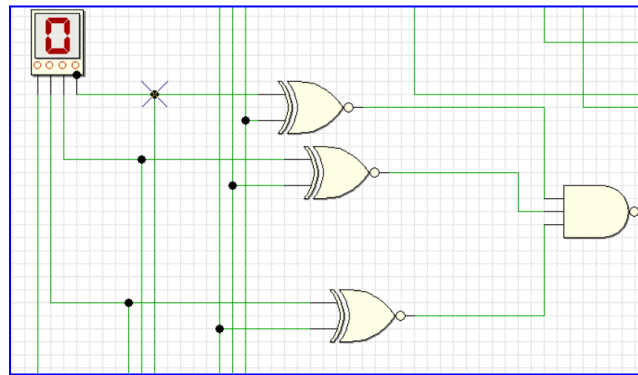
#### 5. 3-bit up/down counter

In our system, we use a 3-bit D positive edge count-up counter. It starts or stops counting depending on the clock enabler. When the 4-bit Arithmetic Comparator (Control up or down comparator) outputs 0, the counter counts up, and when it outputs 1, the counter counts down. The counting stops when the clock pulse is no longer received or when the chosen floor is reached. Additionally, the D Flip-flop is connected to PRE\_bar and CLR\_bar to set or clear the floor display. Each D input is linked to a specific AND gate to control the transition to the next state.



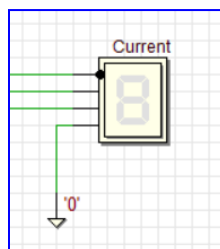
## 6. 3 bit-Comparator

In our system, the comparator uses three XNOR gates and one 3-input NAND gate to compare values from the user's chosen floor input switches and the current floor represented in binary. Each XNOR gate focuses on a different part of the binary numbers, like checking if the least significant bits are the same. If all comparisons match (all three XNOR gates output 1), the NAND gate produces a 0, indicating that the floors are exactly the same and prompting the lift to stop operating. The NAND gate consistently outputs 1 until all comparisons yield the same result.

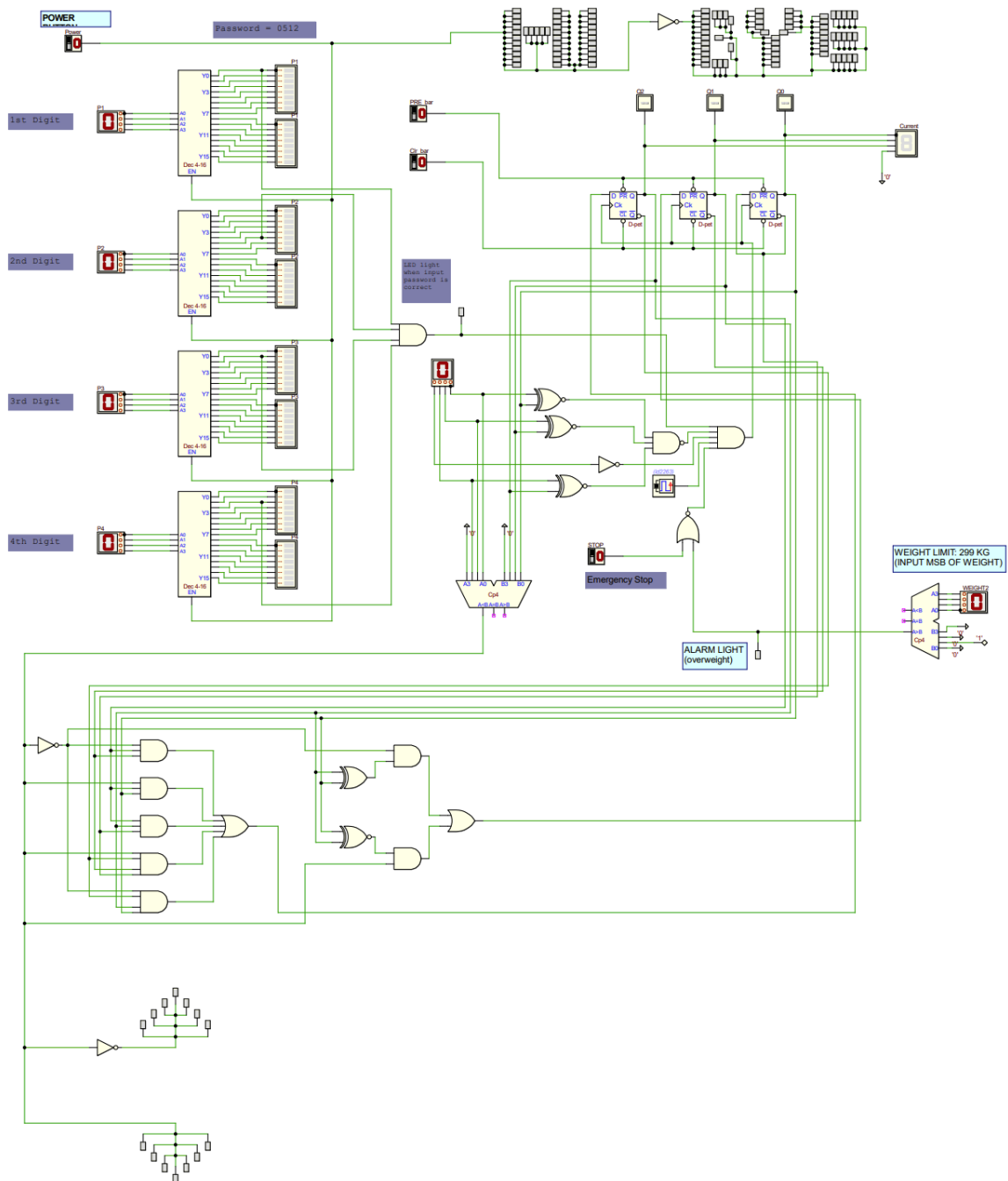


## 7. 7- Segment Display

The 7-Segment display interprets BCD values, converting the received BCD code into decimal numbers and letters for display. This component illuminates the appropriate segments to showcase decimal numbers from 0 to 9 and letters A to F. The 7-Segment display is linked to the current floor value, and as our floor options range from 0 to 7, it effectively displays numbers within this range, ensuring a clear representation of the floor numbers from 0 to 7.



# Full Deeds Drawing



## Conclusion

Throughout the lift electronic controller system project, our achievements have been notable in several key areas. The successful collaboration within the team stands out as a significant accomplishment, as each member brought unique skills, dedication, and perspectives, contributing to the efficient completion of the project. The integration of advanced features, including a 4-digit password, emergency stop, and weight sensor, reflects a commitment to addressing critical aspects such as security, safety, and operational precision. The proficiency demonstrated in circuit design, utilizing combinational and sequential circuits with various components, showcases a strong understanding of digital logic principles.

Our strengths lie in the emphasis on operational precision, ensuring a consistent and accurate arrival at the desired floor, thereby enhancing the overall user experience. Prioritizing safety through features like the emergency stop and password protection system highlights a commitment to passenger well-being and security. The comprehensive design of the lift electronic controller system, meeting basic functionality requirements while incorporating additional features, is another notable strength.

However, there are identified weaknesses in the project. The documentation could be clearer and better organized to enhance overall readability and understanding. Providing a more detailed breakdown of each section and ensuring a logical flow of information would improve the coherence of the report. Additionally, the report briefly mentions future work without providing specific details on areas for improvement or expansion, indicating a need for more detailed planning and proposals for long-term enhancements.

Looking ahead, addressing these weaknesses should be a priority. Future work could include refining the user interface for better interaction, exploring system scalability, incorporating accessibility features, and evaluating energy efficiency measures. By leveraging our strengths and addressing weaknesses, we aim to continually improve the lift electronic controller system, ensuring its effectiveness, security, and adaptability in the long run.

## References

### Internet

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- <https://www.geeksforgeeks.org/magnitude-comparator-in-digital-logic/>
- <https://eevibes.com/digital-logic-design/how-elevator-control-system-works/>

### Book

Digital Logic Fifth Edition ( School of Computing Faculty of Engineering UTM )

## Appendices

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**Video link :**

<https://drive.google.com/file/d/1GyJFtASk4GqO3xNESaOuh3od9udr9PvT/view?usp=sharing>